In the Name of Allah

Signals and Systems

Computer Homework 1

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Student: Seyed Kiarash Eshraghi

Section 1

Part a

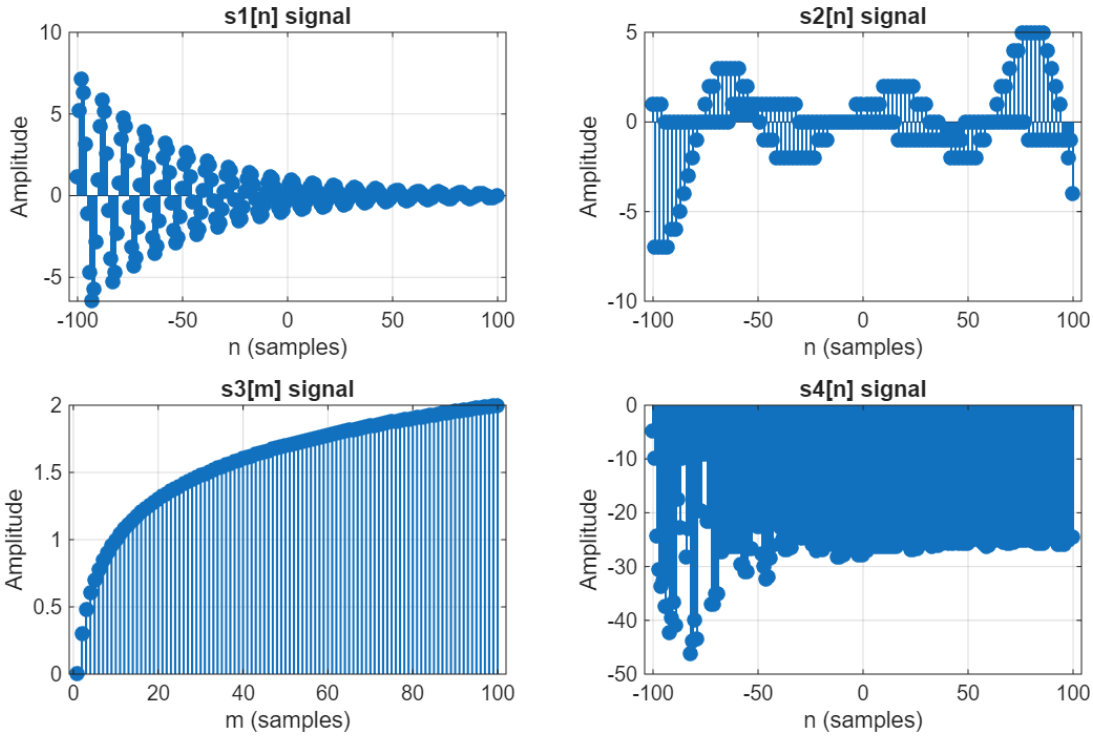


Figure 1: Plots for Section 1, Part a.

Part b

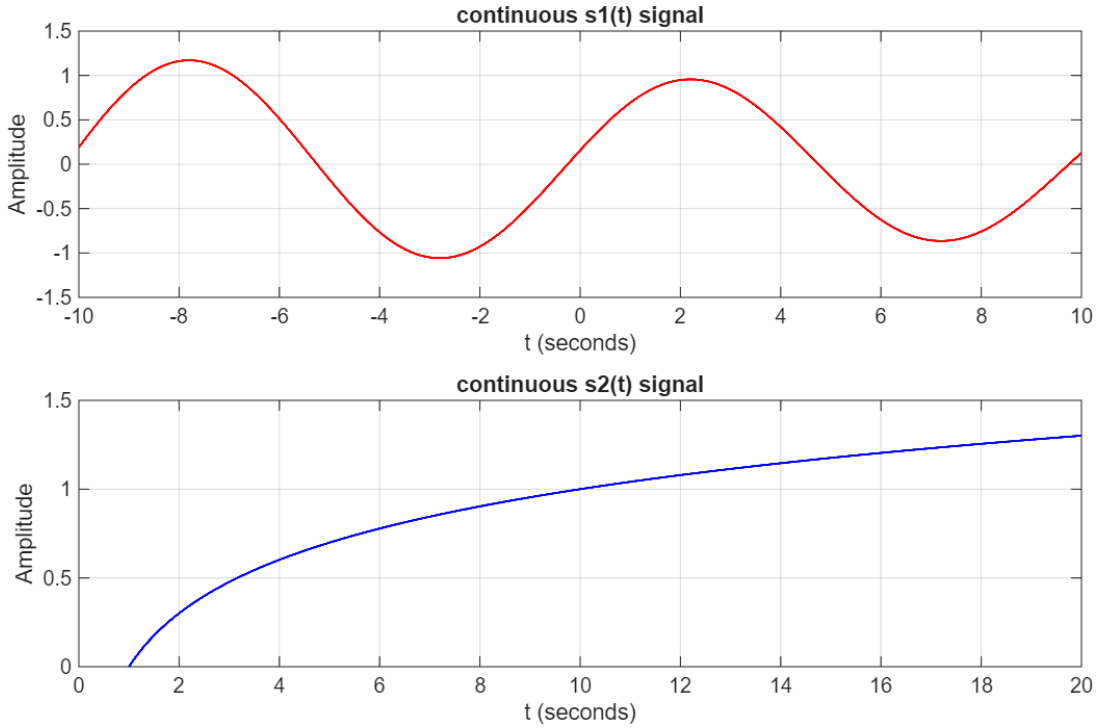


Figure 2: Continuous signals $s_1(t)$ and $s_2(t)$.

Part c

First, the real and imaginary parts of the $s_5(t)$ signal, along with its phase and amplitude, were plotted:

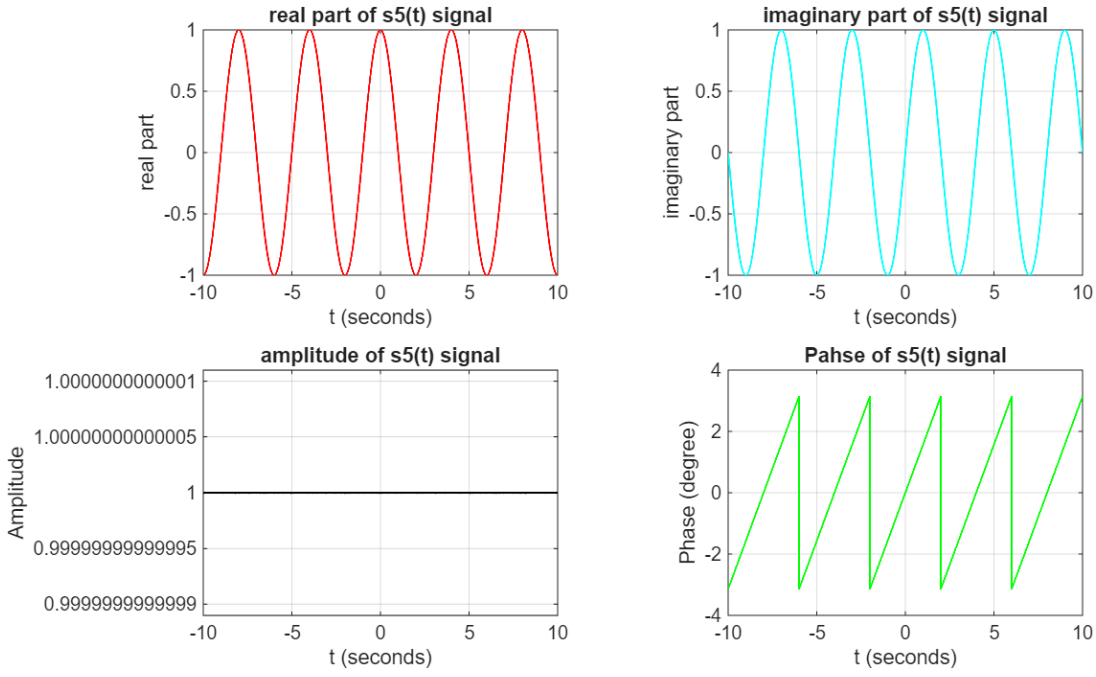


Figure 3: Real part, imaginary part, amplitude, and phase of $s_5(t)$.

The same procedure was applied to the $s_6(t)$ signal:

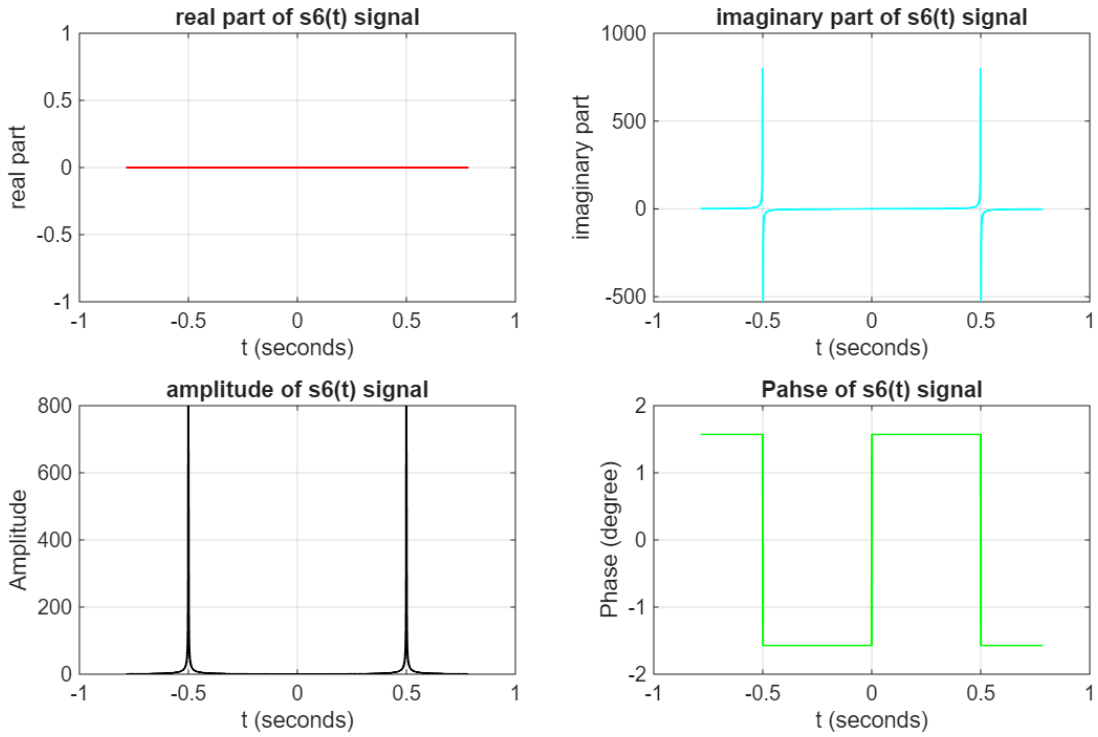


Figure 4: Real part, imaginary part, amplitude, and phase of $s_6(t)$.

Section 2

Part a

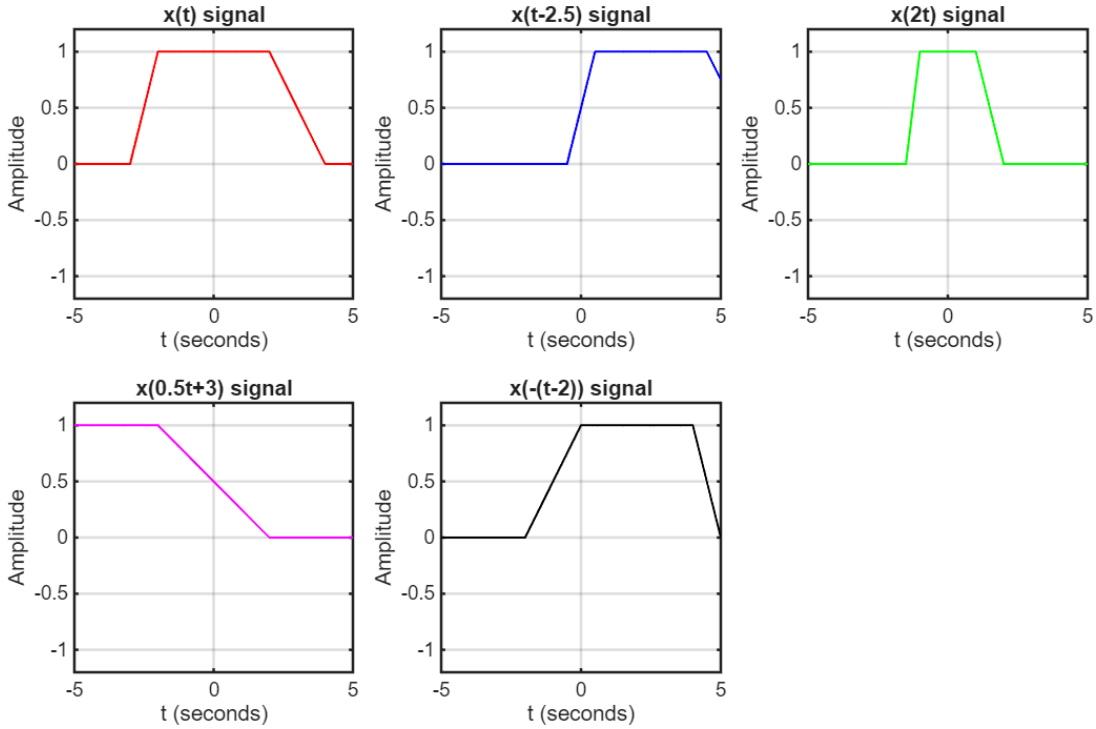


Figure 5: Operations on the $x(t)$ signal.

Part b

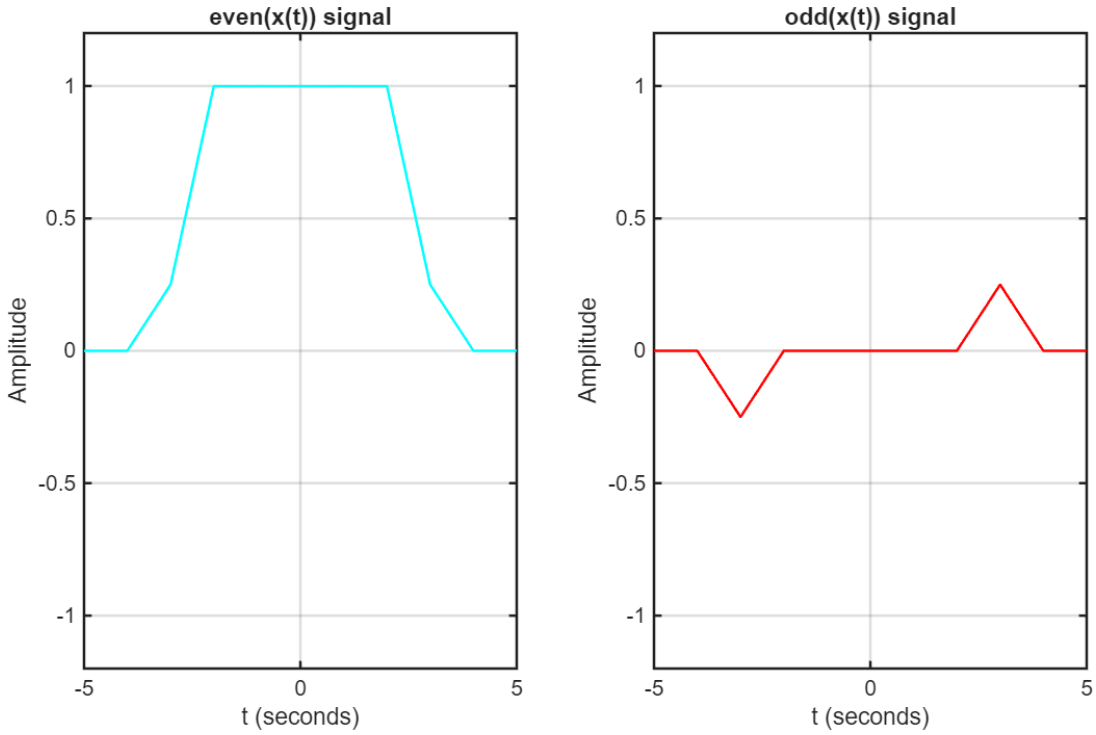


Figure 6: Even and odd parts of $x(t)$.

Section 3

Part a

Two arbitrary signals were chosen to test the custom convolution function:

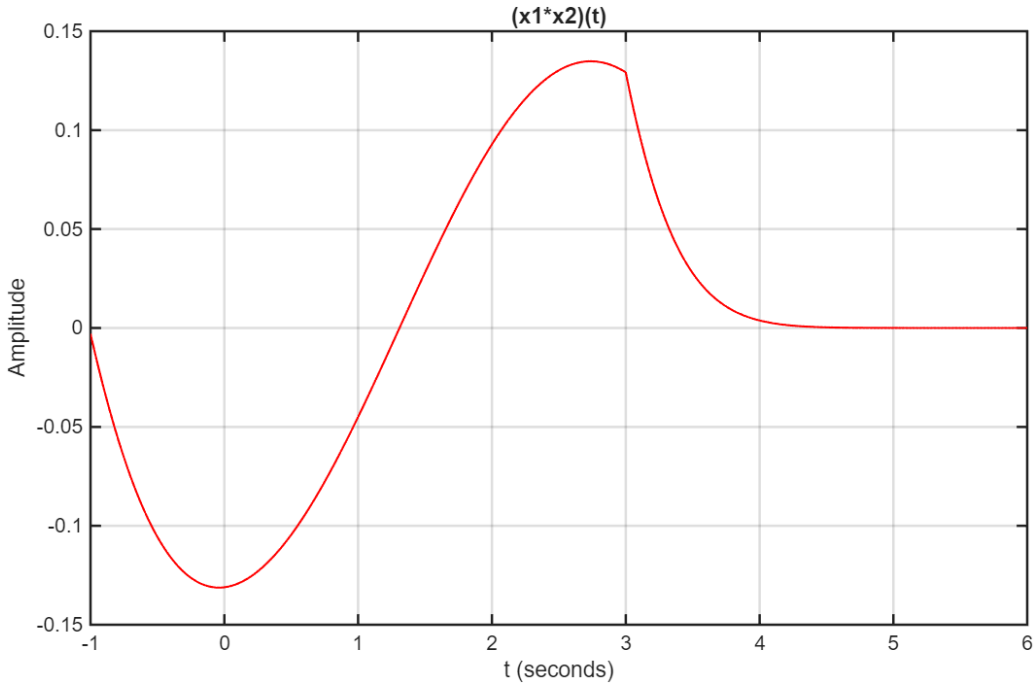
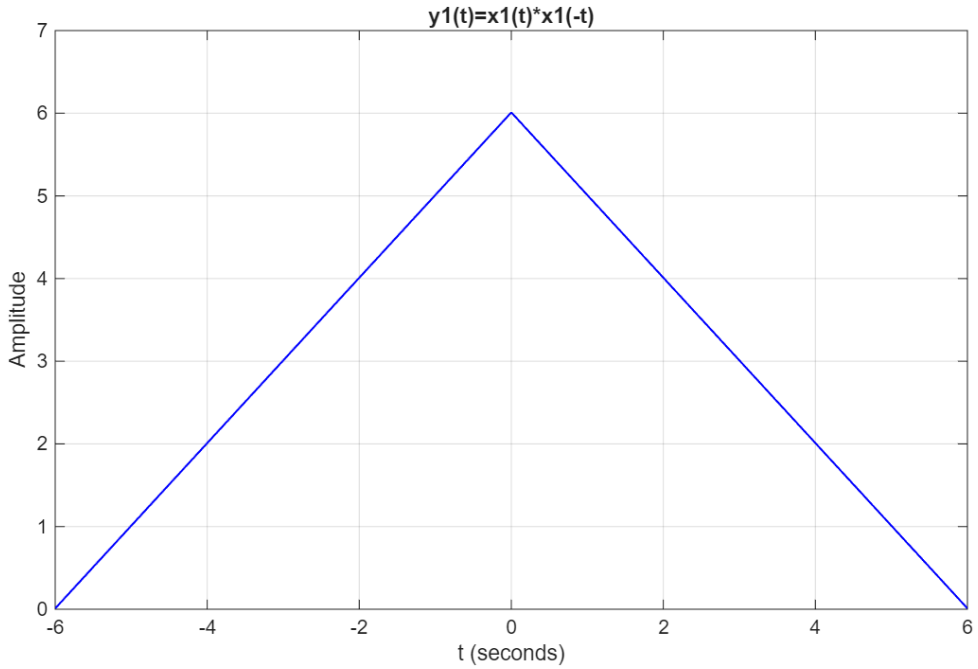
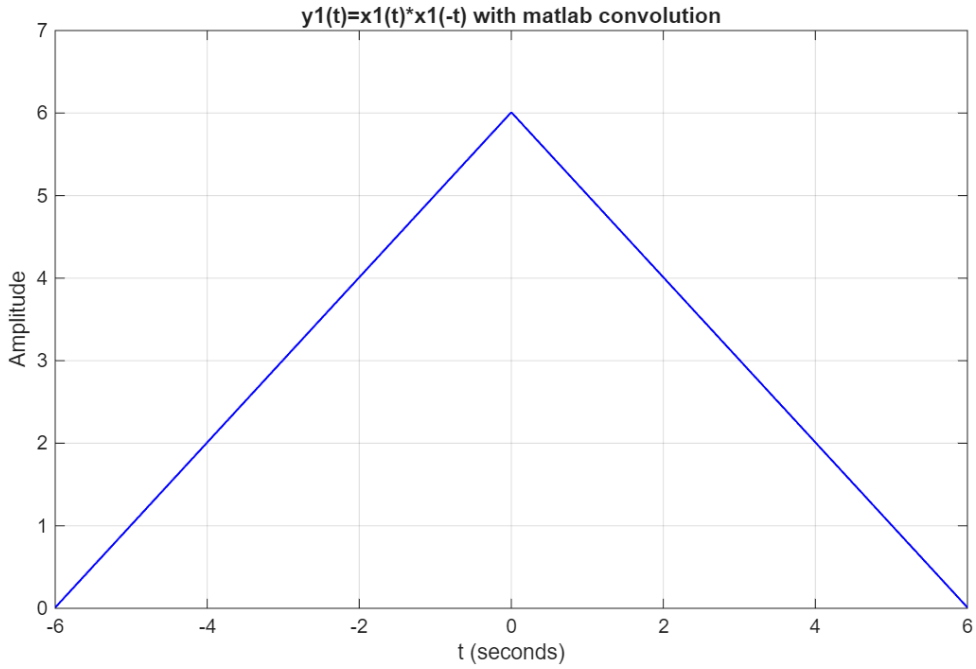
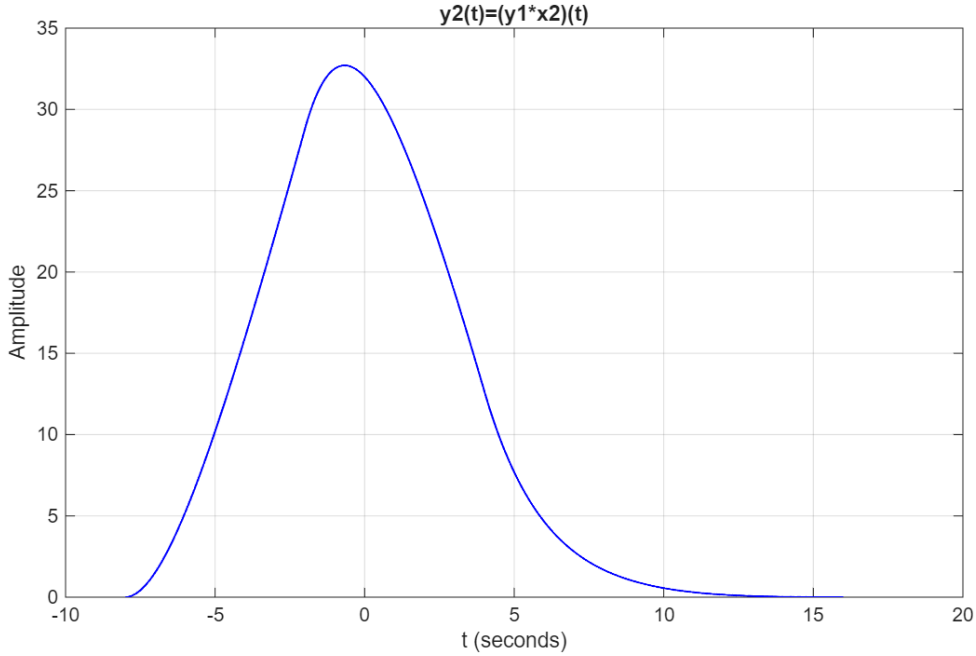


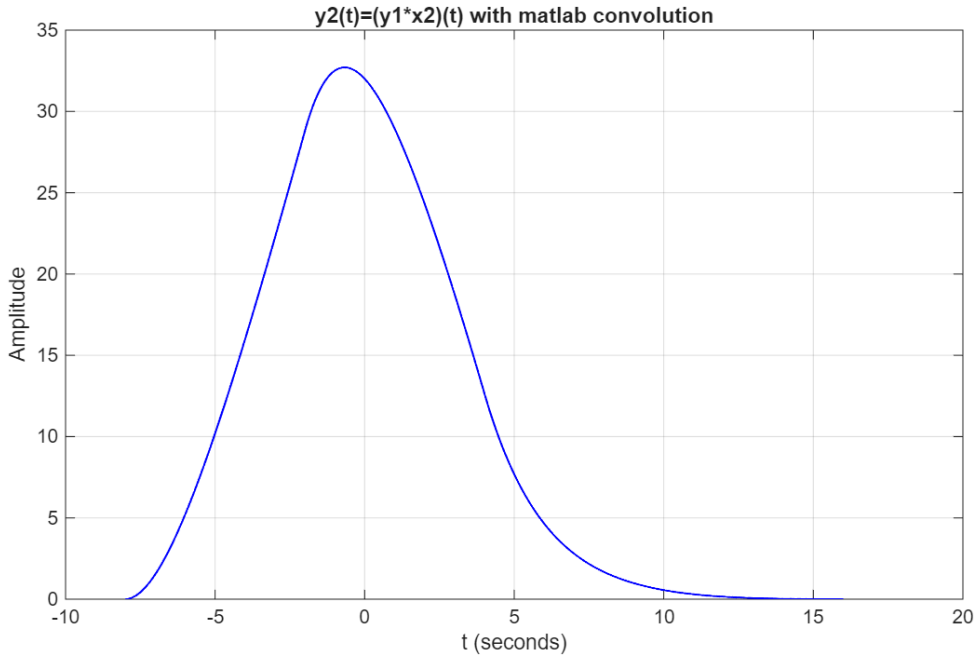
Figure 7: Convolution test using arbitrary signals.

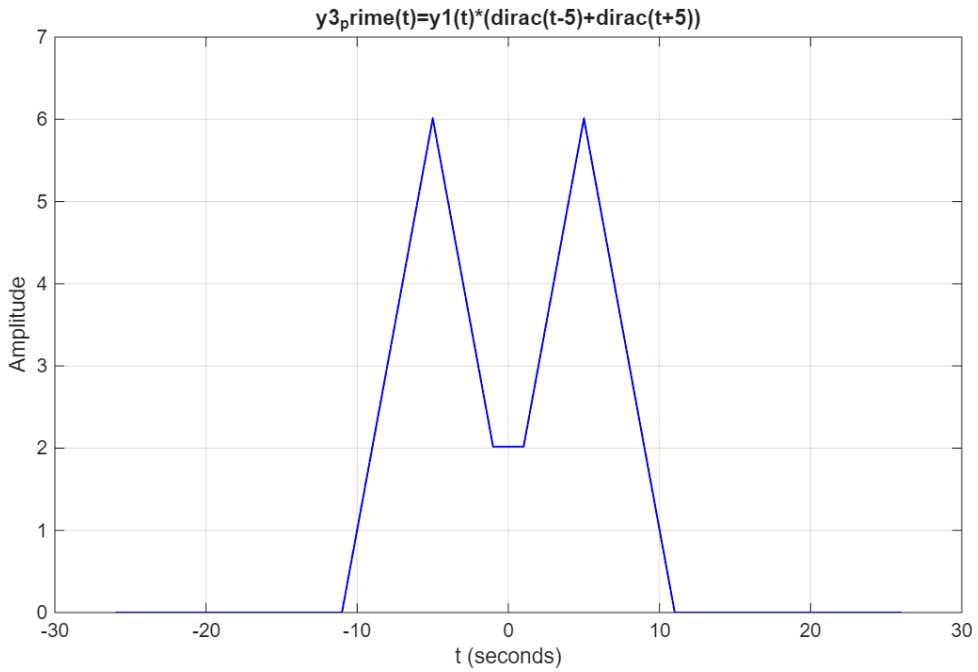
Part b

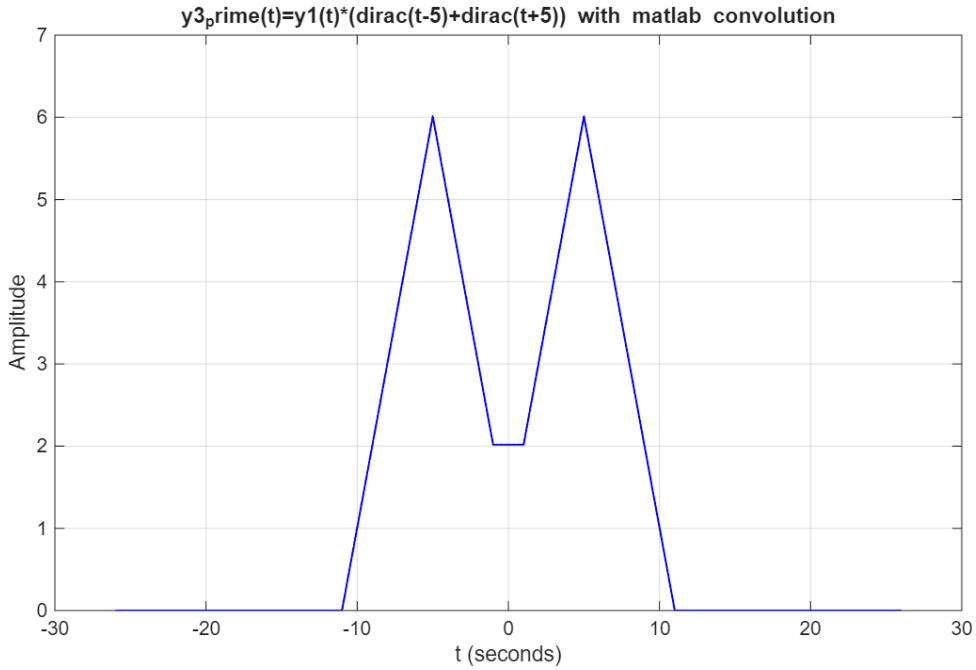


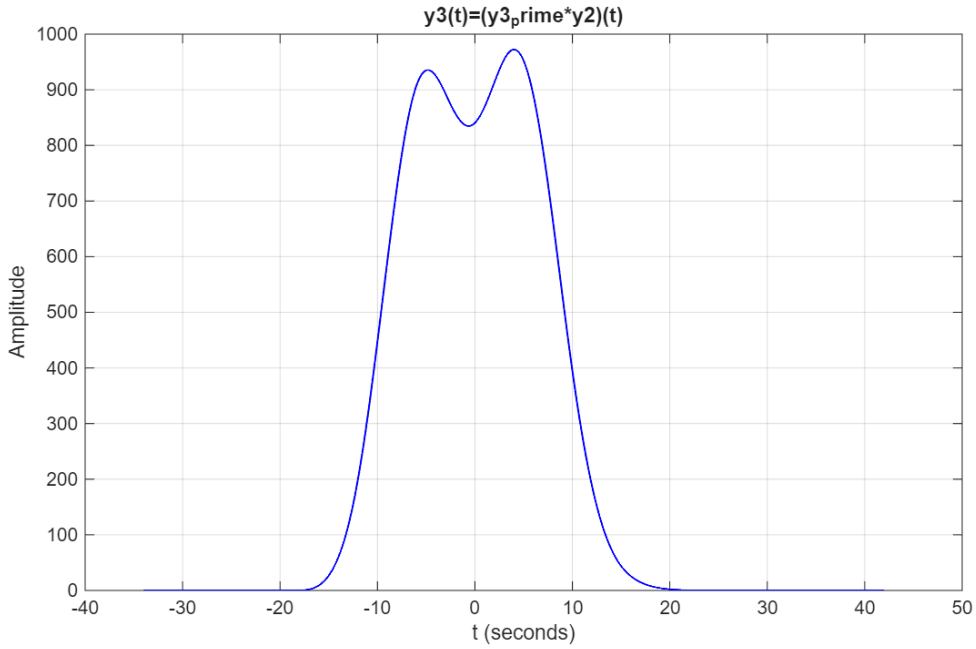

y1(t)=x1(t)*x1(-t) with matlab convolution
Amplitude
t (seconds)
0
1
2
3
4
5
6
7
-6
-4
-2
0
2
4
6

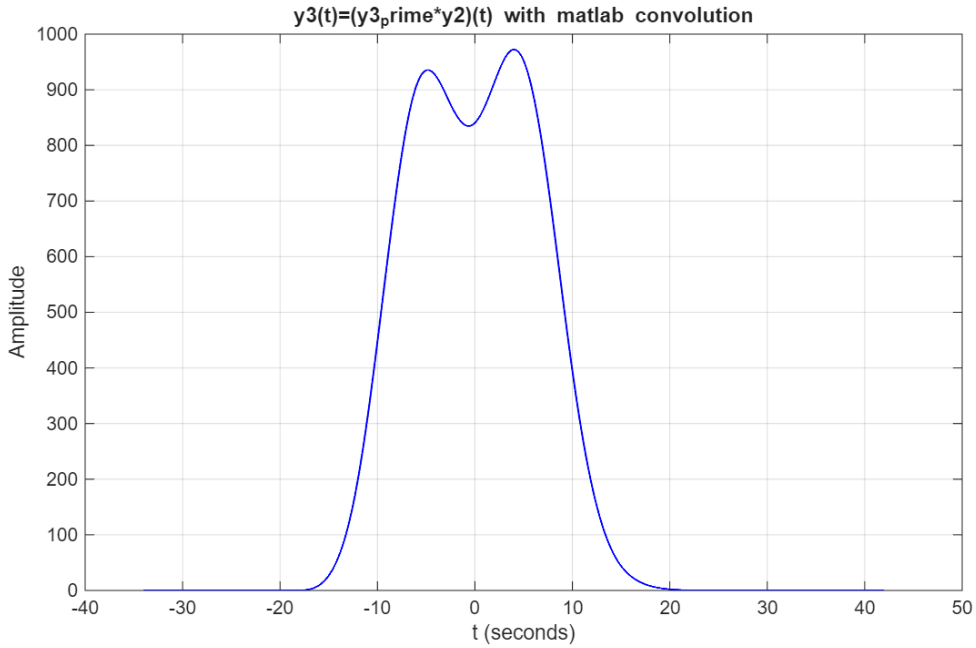

y2(t)=(y1*x2)(t)
Amplitude
t (seconds)
0
5
10
15
20
25
30
35
-10
-5
0
5
10
15
20


y2(t)=(y1*x2)(t) with matlab convolution
Amplitude
t (seconds)
35
30
25
20
15
10
5
0
-10
-5
0
5
10
15
20


y3prime(t)=y1(t)*(dirac(t-5)+dirac(t+5))
Amplitude
t (seconds)
7
6
5
4
3
2
1
0
-30
-20
-10
0
10
20
30

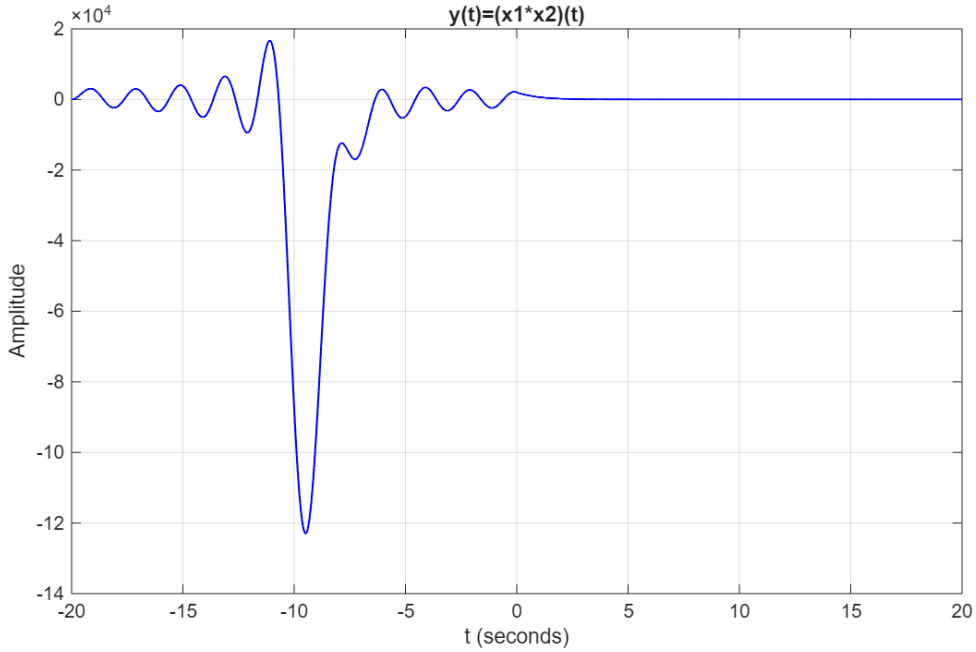

y3prime(t)=y1(t)*(dirac(t-5)+dirac(t+5)) with matlab convolution
Amplitude
t (seconds)
0
1
2
3
4
5
6
7
-30
-20
-10
0
10
20
30

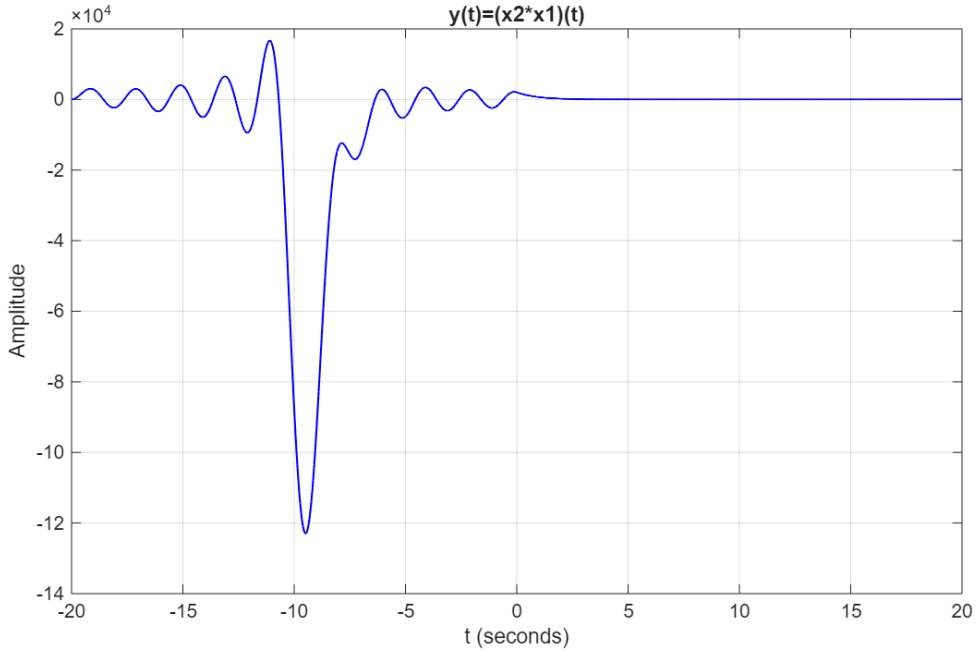

y3(t)=(y3prime*y2)(t)
Amplitude
t (seconds)
0
100
200
300
400
500
600
700
800
900
1000
-40
-30
-20
-10
0
10
20
30
40
50

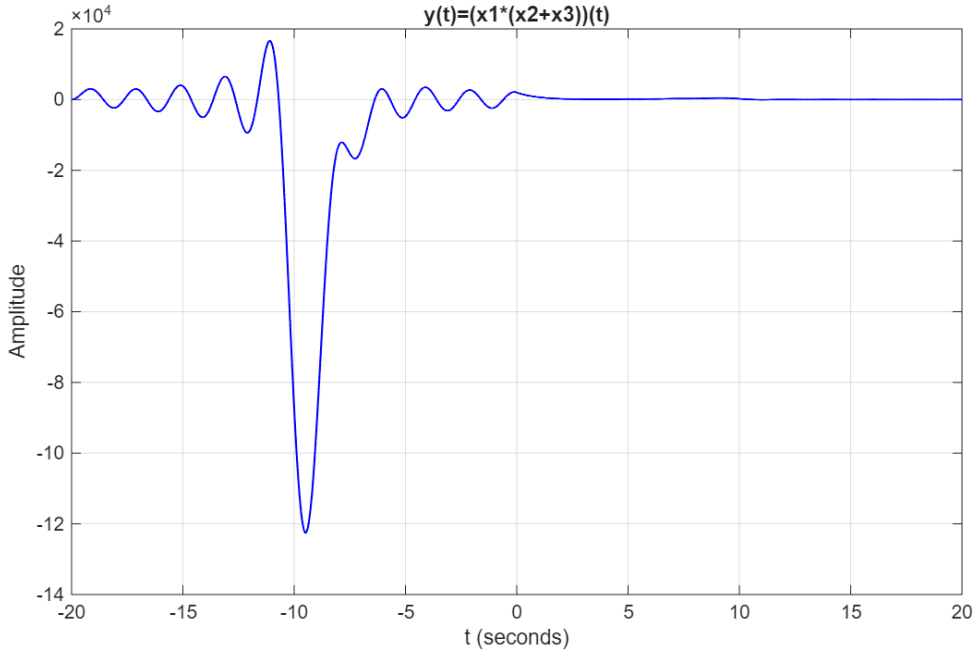


Part c

The resulting plots are provided below, making it straightforward to visually verify the similarity between both sides of the equation:




y(t)=(x2*x1)(t)
×10^4
2
0
-2
-4
-6
-8
-10
-12
-14
Amplitude
-20
-15
-10
-5
0
5
10
15
20
t (seconds)


y(t)=(x1*(x2+x3))(t)
×10^4
2
0
-2
-4
-6
-8
-10
-12
-14
Amplitude
-20
-15
-10
-5
0
5
10
15
20
t (seconds)

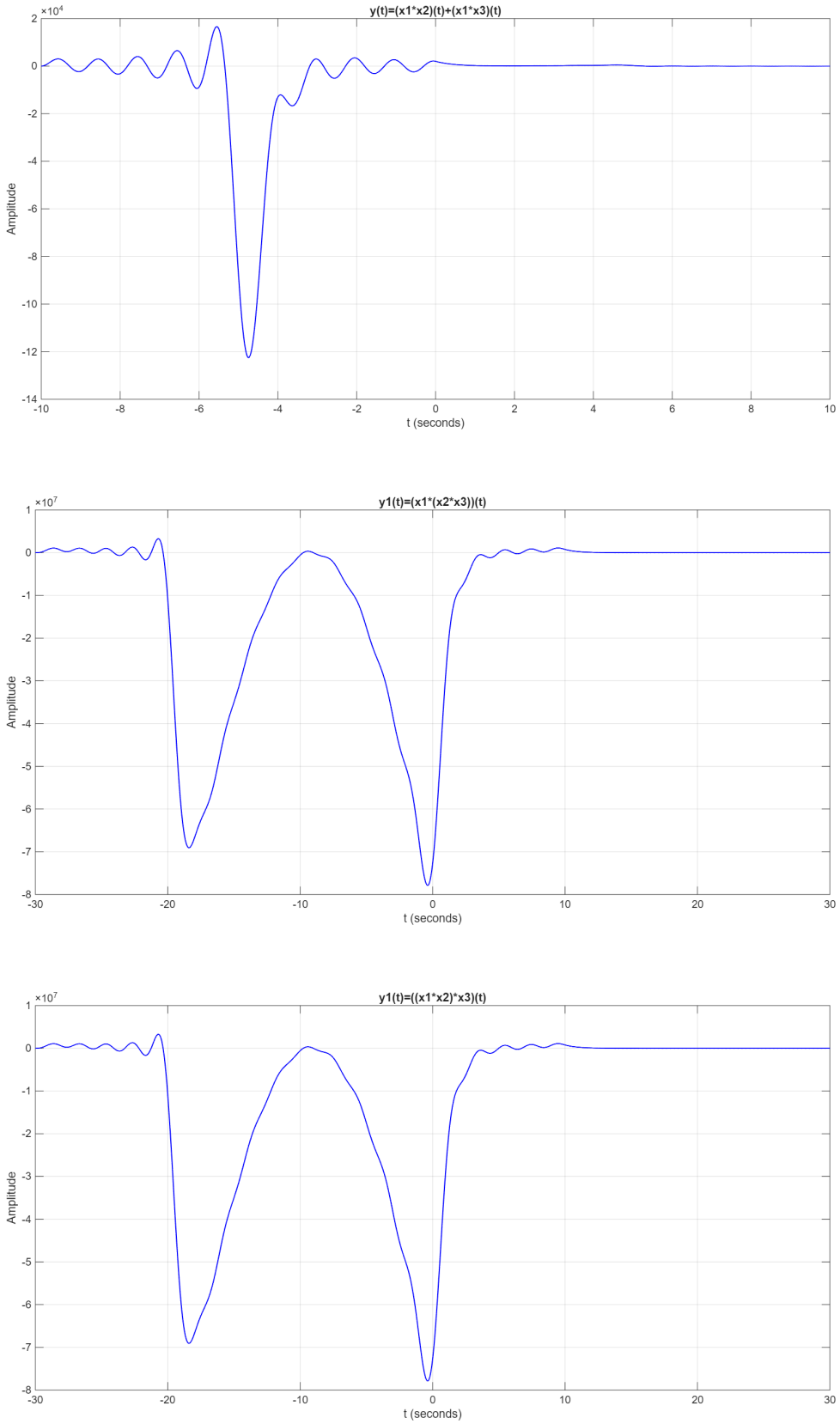


Figure 8: Verification of convolution properties.

Section 4

Part a

The mathematical derivation is as follows:

$$\frac{d}{dt}\left(\Lambda\left(\frac{t}{2}\right)\right) = \begin{cases} 0.5 & -2 < t < 0 \\ -0.5 & 0 < t < 2 \end{cases}$$

$$\frac{d}{dt}(x_1(t)) = -0.5 \times x_2(t)$$

$$x_1(t) = \sum_{k=-\infty}^{\infty} a_k e^{j\pi kt}$$

$$x_2(t) = \sum_{k=-\infty}^{\infty} b_k e^{j\pi kt}$$

$$j\pi k \times a_k = -0.5 \times b_k$$

$$\begin{aligned} 2b_k &= \int_{-1}^{0} -e^{-j\pi kt} dt + \int_{0}^{1} e^{-j\pi kt} dt \\ &= \frac{1}{j\pi k}\left(2 - e^{j\pi k} - e^{-j\pi k}\right) \\ &= \frac{2}{j\pi k}\left(1 - \cos(\pi k)\right) \end{aligned}$$

$$a_k = \frac{0.5}{(\pi k)^2}\left(1 - \cos(\pi k)\right) \quad \text{for } (k \neq 0)$$

$$a_k = 0.75 \quad \text{for } (k = 0)$$

Part b

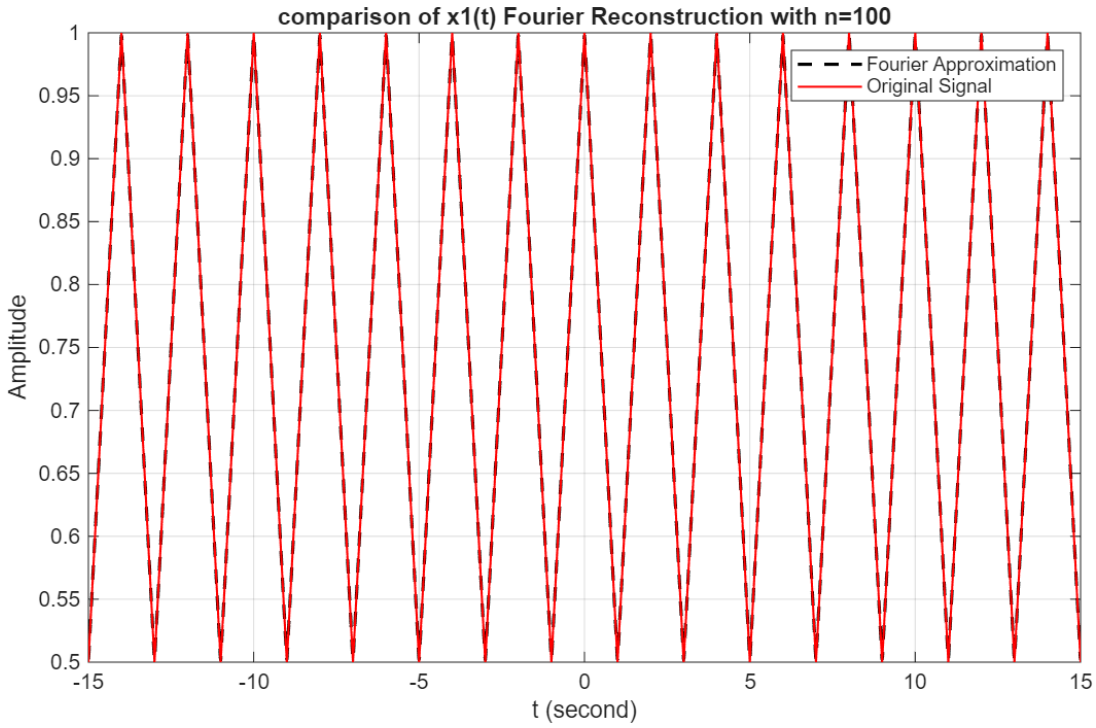


Figure 9: Comparison of $x_1(t)$ Fourier Reconstruction with $n = 100$.

Part c

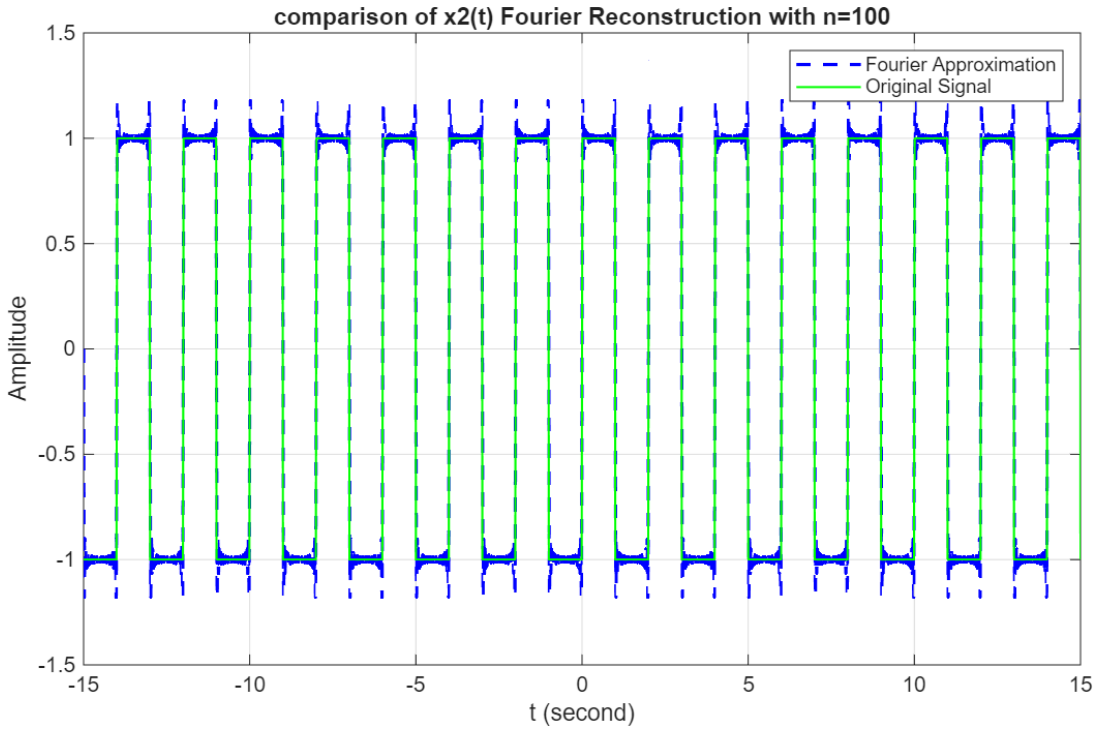


Figure 10: Comparison of $x_2(t)$ Fourier Reconstruction with $n = 100$.

Part d

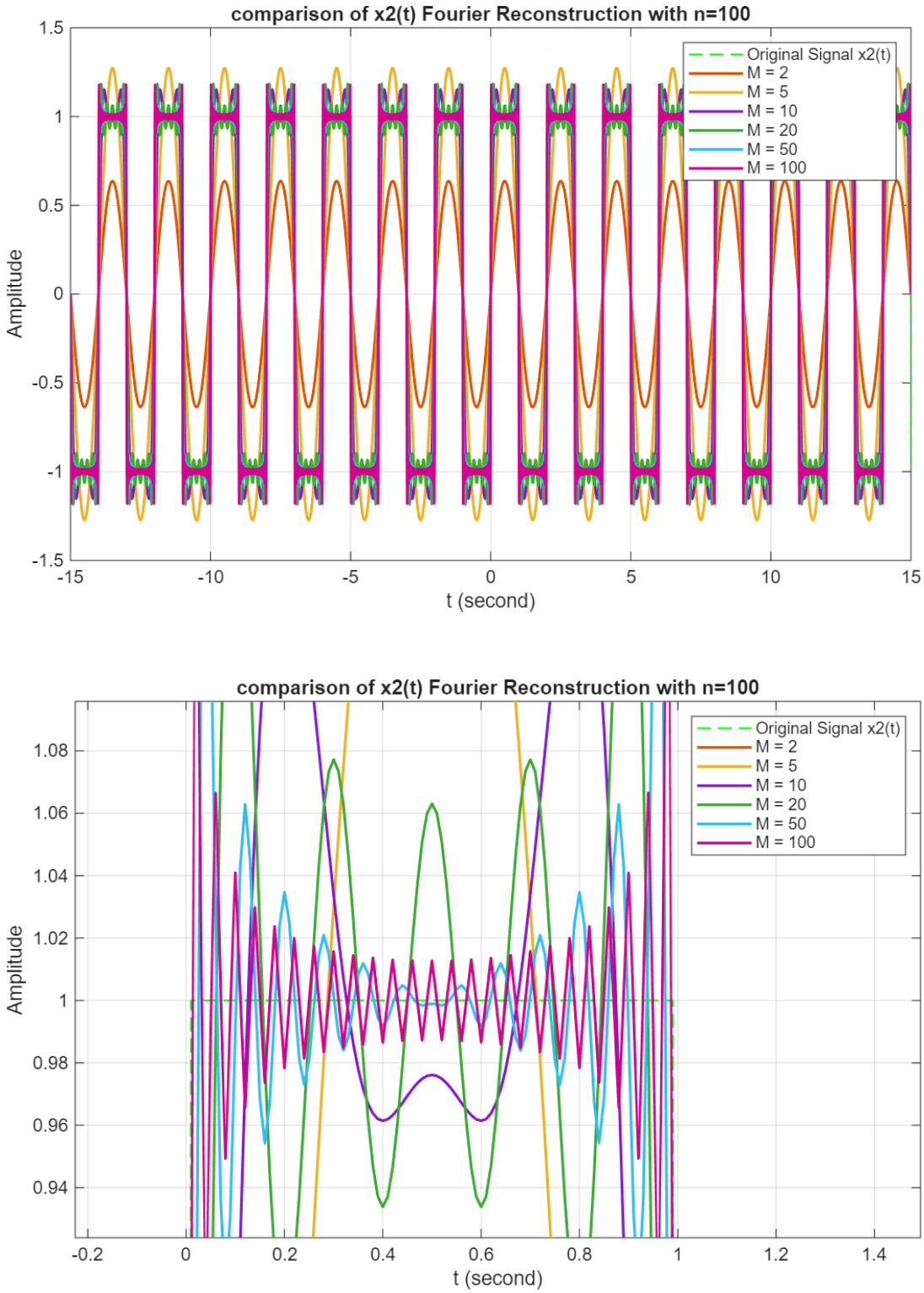


Figure 11: Fourier reconstruction of $x_2(t)$ with varying terms (M).

Part e

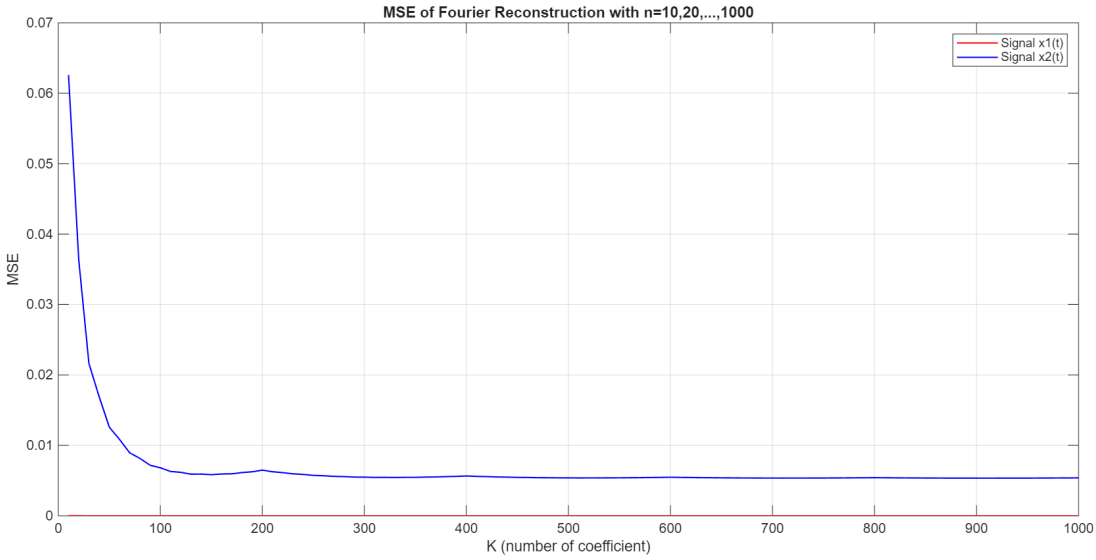


Figure 12: MSE of Fourier Reconstruction with $n = 10, 20, \ldots, 1000$.

Because the definition of $x_2(t)$ at the discontinuity points is set to -1 rather than 0, the Mean Squared Error (MSE) converges to 0.00533 (whereas, theoretically, $x_2(t)$ at the discontinuities should be 0).